RAG Learning

2.2 Equations and Inequalities

OCR GCSE Mathematics (J560) — Paper 1, 2 & 3 — Foundation and Higher

2.2.1 Solving Linear Equations

Linear equation: An equation where the highest power of the unknown is 1. The goal is to find the value of the unknown that makes the equation true.

The golden rule: **whatever you do to one side, do the same to the other side.**

One-Step and Two-Step Equations

Worked Example: Solve $5x - 3 = 17$.

> Add 3 to both sides: $5x = 20$
>
> Divide both sides by 5: $x = 4$

Worked Example: Solve $\dfrac{2x+1}{3} = 5$.

> Multiply both sides by 3: $2x + 1 = 15$
>
> Subtract 1: $2x = 14$
>
> Divide by 2: $x = 7$

Equations with Unknowns on Both Sides

Collect all variable terms onto one side, then all constants onto the other:

Worked Example: Solve $5x + 2 = 3x + 10$.

> Subtract $3x$ from both sides: $2x + 2 = 10$
>
> Subtract 2: $2x = 8$
>
> Divide by 2: $x = 4$

Worked Example: Solve $4(2x - 3) = 6x + 2$.

> Expand: $8x - 12 = 6x + 2$
>
> Subtract $6x$: $2x - 12 = 2$
>
> Add 12: $2x = 14$
>
> Divide by 2: $x = 7$

Worked Example: Solve $\dfrac{x+3}{4} = \dfrac{2x-1}{3}$.

> Multiply through by 12 (LCM of 4 and 3): $3(x + 3) = 4(2x - 1)$
>
> Expand: $3x + 9 = 8x - 4$
>
> Collect terms: $9 + 4 = 8x - 3x$, so $13 = 5x$
>
> $x = \dfrac{13}{5} = 2.6$

Exam Tip: Always check your answer by substituting back into the original equation. This takes 10 seconds and will catch arithmetic slips.

2.2.2 Forming and Solving Equations

Many real-world and geometry problems require you to write an equation before solving it.

Worked Example: The angles of a triangle are $3x$, $(2x+10)^\circ$, and $(x-4)^\circ$. Find x and the three angles.

> Angles in a triangle sum to 180°: $3x + (2x+10) + (x-4) = 180$
>
> Simplify: $6x + 6 = 180$
>
> $6x = 174$, so $x = 29$
>
> Angles: 87°, 68°, 25°. Check: $87 + 68 + 25 = 180$ ✓

Worked Example: A rectangle has length $(3x+2)$ cm and width $(x-1)$ cm. Its perimeter is 46 cm. Find its area.

> Perimeter: $2(3x+2) + 2(x-1) = 46$
>
> $6x + 4 + 2x - 2 = 46$
>
> $8x + 2 = 46 \Rightarrow 8x = 44 \Rightarrow x = 5.5$
>
> Length $= 3(5.5) + 2 = 18.5$ cm, Width $= 4.5$ cm
>
> Area $= 18.5 \times 4.5 = 83.25$ cm²

Worked Example: A bag of sweets contains n sweets. Ali takes 5, then Ben takes half of what remains. 12 sweets are left. How many sweets were in the bag?

> After Ali: $n - 5$
>
> After Ben: $\dfrac{n-5}{2} = 12$
>
> $n - 5 = 24 \Rightarrow n = 29$

2.2.3 Linear Inequalities

An **inequality** shows that one expression is greater than (or less than) another. The solution is a range of values, not a single value.

Symbols: $<$ (less than), $\leq$ (less than or equal to), $>$ (greater than), $\geq$ (greater than or equal to).

Solving Linear Inequalities

Use the same inverse operations as equations, with one critical difference:

When you multiply or divide both sides by a negative number, you must reverse the inequality sign.

Worked Example: Solve $3x - 7 > 5$.

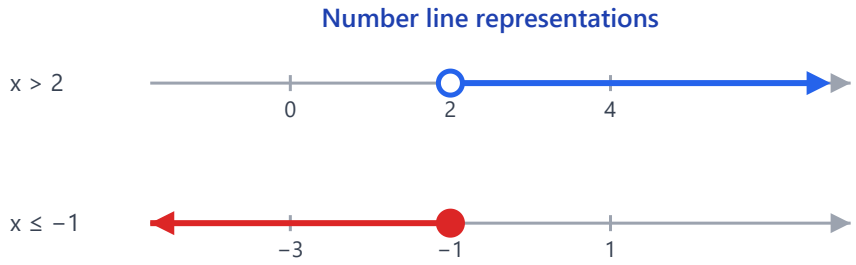
> Add 7: $3x > 12$
>
> Divide by 3: $x > 4$

Worked Example: Solve $-2x + 1 \leq 9$.

> Subtract 1: $-2x \leq 8$
>
> Divide by -2 (reverse sign!): $x \geq -4$

Representing Inequalities on a Number Line

Use an **open circle** ○ for strict inequalities ($<$ or $>$) and a **filled circle** ● for $\leq$ or $\geq$.



Combined Inequalities

E.g. $-3 < 2x + 1 \leq 7$: solve as two separate inequalities or work on all three parts simultaneously.

Worked Example: Solve $-3 < 2x + 1 \leq 7$ and represent on a number line.

> Subtract 1 from all three parts: $-4 < 2x \leq 6$
>
> Divide all three parts by 2: $-2 < x \leq 3$
>
> Integer values: $x \in \{-1, 0, 1, 2, 3\}$
>
> Number line: open circle at -2, filled circle at 3, line between them.

Exam Tip: When listing integer solutions, make sure to include the endpoints only if the sign is $\leq$ or $\geq$.

2.2.4 Simultaneous Equations

Two equations in two unknowns — find values of both unknowns that satisfy *both* equations simultaneously.

Method 1: Elimination

Add or subtract multiples of the equations to eliminate one variable.

Worked Example: Solve: $3x + 2y = 13$ and $5x - 2y = 11$.

Coefficients of y are $+2$ and -2, so add the equations:

$8x = 24 \Rightarrow x = 3$

Substitute into equation 1: $9 + 2y = 13 \Rightarrow 2y = 4 \Rightarrow y = 2$

Check in equation 2: $15 - 4 = 11$ ✓

Worked Example: Solve: $2x + 3y = 16$ and $5x - y = 2$.

Multiply equation 2 by 3: $15x - 3y = 6$

Add to equation 1: $17x = 22 \Rightarrow x = \dfrac{22}{17}$

Or — to get integers here, multiply eq 1 by 5 and eq 2 by 2: $10x + 15y = 80$ and $10x - 2y = 4$. Subtract: $17y = 76 \Rightarrow y = \dfrac{76}{17}$. Messy — but systematic.

Better: multiply eq 2 by 3 to match y coefficient: $15x - 3y = 6$. Add to eq 1: $17x = 22$, so $x = \frac{22}{17}$, and back-substitute for y.

Method 2: Substitution

Rearrange one equation to express one variable in terms of the other, then substitute into the second equation.

Worked Example: Solve: $y = 2x - 1$ and $3x + 4y = 18$.

Substitute $y = 2x - 1$ into the second equation:

$3x + 4(2x - 1) = 18$

$3x + 8x - 4 = 18 \Rightarrow 11x = 22 \Rightarrow x = 2$

$y = 2(2) - 1 = 3$

Solution: $x = 2,\ y = 3$

Graphical Method

Plot both equations as straight lines. The solution is the point of intersection of the two lines.

- Parallel lines (same gradient, different intercept) → no solution.
- Coincident lines (same equation) → infinitely many solutions.

> **Exam Tip:** In elimination, make sure the signs are correct when subtracting — this is the most common error. Write the subtraction out explicitly.

2.2.5 Quadratic Equations

A quadratic equation has the form $ax^2 + bx + c = 0$ and generally has **two solutions** (roots).

Method 1: Factorising

Rearrange to equal zero, factorise, then set each factor equal to zero:

> **Worked Example:** Solve $x^2 - 5x + 6 = 0$.
>
> > Factorise: $(x-2)(x-3) = 0$
> >
> > Either $x - 2 = 0$ or $x - 3 = 0$
> >
> > $x = 2$ or $x = 3$

> **Worked Example:** Solve $2x^2 + x - 6 = 0$.
>
> > Factorise (AC method: $ac = -12$; factors -3 and 4): $(2x-3)(x+2) = 0$
> >
> > $x = \frac{3}{2}$ or $x = -2$

Method 2: The Quadratic Formula

For any quadratic $ax^2 + bx + c = 0$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The expression $b^2 - 4ac$ is called the **discriminant**:

Value of $b^2 - 4ac$	Number of real roots	Nature of roots
$b^2 - 4ac > 0$	2 distinct real roots	Two different solutions
$b^2 - 4ac = 0$	1 repeated root	One solution (parabola touches x-axis)
$b^2 - 4ac < 0$	No real roots	Parabola does not cross x-axis

Worked Example: Solve $3x^2 - 5x - 2 = 0$ using the quadratic formula.

> $a = 3,\ b = -5,\ c = -2$
>
> $b^2 - 4ac = 25 - 4(3)(-2) = 25 + 24 = 49$
>
> $x = \frac{5 \pm \sqrt{49}}{6} = \frac{5 \pm 7}{6}$
>
> $x = \frac{12}{6} = 2$ or $x = \frac{-2}{6} = -\frac{1}{3}$

Worked Example: Solve $x^2 + 3x - 1 = 0$, giving answers to 2 decimal places.

> $a = 1,\ b = 3,\ c = -1$
>
> $b^2 - 4ac = 9 + 4 = 13$
>
> $x = \frac{-3 \pm \sqrt{13}}{2}$
>
> $x = \frac{-3 + 3.6056}{2} = 0.30$ (2 d.p.) or $x = \frac{-3 - 3.6056}{2} = -3.30$ (2 d.p.)

Exam Tip: The quadratic formula always works, even when factorising seems impossible. Always rearrange to $ax^2 + bx + c = 0$ first. Identify a, b, c before substituting — include the

sign.

Higher — Completing the Square

Write $ax^2 + bx + c$ in the form $a(x+p)^2 + q$. This reveals the vertex of the parabola and is a third method for solving quadratics.

For $x^2 + bx + c$ (coefficient of x^2 is 1):

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2 + c$$

Worked Example: Solve $x^2 + 6x + 2 = 0$ by completing the square.

> $(x+3)^2 - 9 + 2 = 0$
> $(x+3)^2 = 7$
> $x + 3 = \pm\sqrt{7}$
> $x = -3 \pm \sqrt{7}$

Worked Example: Write $2x^2 - 8x + 5$ in the form $a(x+p)^2 + q$. Hence state the minimum value and the x-coordinate where it occurs.

> Factor out 2: $2(x^2 - 4x) + 5$
> Complete the square inside: $2\left[(x-2)^2 - 4\right] + 5$
> Expand: $2(x-2)^2 - 8 + 5 = 2(x-2)^2 - 3$
> Minimum value is -3 at $x = 2$

2.2.6 Algebraic Proof [Higher]

Higher only

Algebraic proof uses algebra to show that a statement is always true. You must work with general expressions, not specific numbers.

Key vocabulary:

- Even number: $2n$
- Odd number: $2n + 1$
- Consecutive integers: $n, n + 1, n + 2, \ldots$
- Multiple of k: kn (where n is any integer)

Showing Identities

Worked Example: Prove that $(n + 3)^2 - (n + 1)^2 \equiv 4(n + 2)$.

> LHS: $(n + 3)^2 - (n + 1)^2$
> $= n^2 + 6n + 9 - (n^2 + 2n + 1)$
> $= n^2 + 6n + 9 - n^2 - 2n - 1$
> $= 4n + 8 = 4(n + 2)$ = RHS ✓

Proving Divisibility

Worked Example: Prove that the sum of any three consecutive integers is always divisible by 3.

> Let the three integers be n, $n + 1$, $n + 2$
> Sum $= n + (n + 1) + (n + 2) = 3n + 3 = 3(n + 1)$
> This is a multiple of 3, so the sum is always divisible by 3. □

Worked Example: Prove that the product of two odd numbers is always odd.

> Let the two odd numbers be $2m + 1$ and $2n + 1$
>
> Product $= (2m + 1)(2n + 1) = 4mn + 2m + 2n + 1 = 2(2mn + m + n) + 1$
>
> This is in the form $2k + 1$ (where $k = 2mn + m + n$ is an integer), so the product is odd. $\square$

Disproving by Counterexample

To disprove a general claim, you only need to find ONE example where it fails — a **counterexample**.

Worked Example: Disprove: "The square of any odd number is always 1 more than a multiple of 8."

> Wait — actually this is TRUE. $(2n + 1)^2 = 4n^2 + 4n + 1 = 4n(n + 1) + 1$. Since one of n or $n + 1$ is even, $4n(n + 1)$ is a multiple of 8, so $(2n + 1)^2 \equiv 1 \pmod 8$.
>
> Alternative disproof example — "The sum of two prime numbers is always even": counterexample is $2 + 3 = 5$ (odd). $\square$

Exam Tip: In a proof question, use fully general algebraic expressions. Do not use specific numbers. End with a clear concluding sentence explaining what you have shown.

Quick Reference Summary

Topic	Key method / formula

Linear equations	Inverse operations; same operation on both sides
Inequalities	Same as equations but reverse sign when multiplying/dividing by a negative
Simultaneous (linear)	Elimination (match coefficients) or substitution
Quadratic: factorising	$(x-p)(x-q)=0 \Rightarrow x=p$ or $x=q$
Quadratic formula	$x=\dfrac{-b\pm\sqrt{b^2-4ac}}{2a}$
Completing the square [H]	$x^2+bx+c=(x+\frac{b}{2})^2-(\frac{b}{2})^2+c$
Algebraic proof [H]	Use $2n$ (even), $2n+1$ (odd), $n, n+1, \ldots$ (consecutive)

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